Then, for any $i \in [M]$ we have

$$\mathop{\mathbb{E}}_{\tau}\Big[\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}} \end{pmatrix}\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \\ y_{\tau,i}^{\text{sfx}} \end{pmatrix}^{\top}\Big] = \mathop{\mathbb{E}}_{w_\tau, x_{\tau,i}^{\text{sfx}}}\begin{pmatrix} x_{\tau,i}^{\text{sfx}} \cdot (x_{\tau,i}^{\text{sfx}})^{\top} & x_{\tau,i}^{\text{sfx}} \cdot (w_\tau^{\top} x_{\tau,i}^{\text{sfx}})^{\top} \\ w_\tau^{\top} x_{\tau,i}^{\text{sfx}} \cdot (x_{\tau,i}^{\text{sfx}})^{\top} & w_\tau^{\top} x_{\tau,i}^{\text{sfx}} \cdot (w_\tau^{\top} x_{\tau,i}^{\text{sfx}})^{\top} \end{pmatrix}$$

$$= \begin{pmatrix} \Lambda & \Lambda \cdot 0_{d\times 1} \\ 0_{1\times d} \cdot \Lambda & \mathbb{E}_{w_\tau}\Big[w_\tau^{\top}\Lambda w_\tau\Big] \end{pmatrix} = \begin{pmatrix} \Lambda & 0_{d\times 1} \\ 0_{1\times d} & \underbrace{\text{Tr}(I_d\Lambda)}_{\text{by Lemma A.2}} \end{pmatrix} = \begin{pmatrix} \Lambda & 0_{d\times 1} \\ 0_{1\times d} & \text{Tr}(\Lambda) \end{pmatrix}. \tag{A.13}$$

Finally, by inserting Eq. (A.13) into Eq. (A.12), $\ell_3(\theta)$ can thus be simplified as follows,

$$\ell_3(\theta) = \frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau}\Big[\|W_{11}^{KQ} x_{\tau,q}\|_2^2\Big] \cdot \begin{pmatrix} (w_{21}^V)^{\top} & w_{22}^V \end{pmatrix} \cdot \left(\sum_{i=1}^{M_{\text{train}}} \begin{pmatrix} \Lambda & 0_{d\times 1} \\ 0_{1\times d} & \text{Tr}(\Lambda) \end{pmatrix}\right) \cdot \begin{pmatrix} (w_{21}^V)^{\top} & w_{22}^V \end{pmatrix}^{\top}$$

$$= \frac{2\epsilon^2 M_{\text{train}}^2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau}\Big[\|W_{11}^{KQ} x_{\tau,q}\|_2^2\Big] \cdot \Big((w_{21}^V)^{\top}\Lambda w_{21}^V + \text{Tr}(\Lambda)(w_{22}^V)^2\Big). \tag{A.14}$$

According to Eq. (A.14), $\ell_3(\theta)$ does not depend on w_{21}^{KQ}, which means that $\partial_{w_{21}^{KQ}}\ell_3(\theta) = 0_{1\times d}$. On the other hand, based on Eq. (A.14), when $w_{21}^V = 0$, the derivative of $\ell_3(\theta)$ with respect to w_{21}^V is calculated as follows,

$$\partial_{w_{21}^V}\ell_3(\theta)\Big|_{w_{21}^V=0} = \partial_{w_{21}^V}\Big[\frac{2\epsilon^2 M_{\text{train}}^2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau}\Big[\|W_{11}^{KQ} x_{\tau,q}\|_2^2\Big] \cdot \Big((w_{21}^V)^{\top}\Lambda w_{21}^V + \text{Tr}(\Lambda)(w_{22}^V)^2\Big)\Big]\Big|_{w_{21}^V=0}$$

$$= \frac{2\epsilon^2 M_{\text{train}}^2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau}\Big[\|W_{11}^{KQ} x_{\tau,q}\|_2^2\Big] \cdot \partial_{w_{21}^V}\Big[(w_{21}^V)^{\top}\Lambda w_{21}^V\Big]\Big|_{w_{21}^V=0}$$

$$= \frac{4\epsilon^2 M_{\text{train}}^2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau}\Big[\|W_{11}^{KQ} x_{\tau,q}\|_2^2\Big] \cdot \Big[(w_{21}^V)^{\top}\Lambda\Big]\Big|_{w_{21}^V=0}$$

$$= \frac{4\epsilon^2 M_{\text{train}}^2}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau}\Big[\|W_{11}^{KQ} x_{\tau,q}\|_2^2\Big] \cdot 0_{d\times 1}^{\top}\Lambda = 0_{1\times d},$$

which justifies our claim in Step 3.

Step 4: Show that $w_{21}^V = w_{21}^{KQ} = 0_{d\times 1}$ indicates $\partial_{w_{21}^V}\ell_4(\theta) = \partial_{w_{21}^{KQ}}\ell_4(\theta) = 0_{1\times d}$. When $w_{21}^V = w_{21}^{KQ} = 0_{d\times 1}$, based on the expression of $\ell_4(\theta)$ given in Eq. (A.11), the derivative of $\ell_4(\theta)$ with respect to w_{21}^V is calculated as follows,

$$\partial_{w_{21}^V}\ell_4(\theta)\Big|_{w_{21}^V=w_{21}^{KQ}=0_{d\times 1}} = \partial_{w_{21}^V}\Big[\frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2}\|w_{21}^V\|_2^2 \cdot \mathop{\mathbb{E}}_{\tau}\Big\|\begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix}^{\top}\begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^{\top} \end{pmatrix} x_{\tau,q}\Big\|_2^2\Big]\Big|_{w_{21}^V=w_{21}^{KQ}=0_{d\times 1}}$$

$$= \Big[\frac{4\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau}\Big\|\begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix}^{\top}\begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^{\top} \end{pmatrix} x_{\tau,q}\Big\|_2^2 \cdot (w_{21}^V)^{\top}\Big]\Big|_{w_{21}^V=w_{21}^{KQ}=0_{d\times 1}}$$

$$= \frac{4\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot \mathop{\mathbb{E}}_{\tau}\Big\|\begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix}^{\top}\begin{pmatrix} W_{11}^{KQ} \\ 0_{d\times 1}^{\top} \end{pmatrix} x_{\tau,q}\Big\|_2^2 \cdot 0_{d\times 1}^{\top} = 0_{1\times d}.$$

Besides, for the derivative of $\ell_4(\theta)$ with respect to w_{21}^{KQ}, we also have that

$$\partial_{w_{21}^{KQ}}\ell_4(\theta)\Big|_{w_{21}^V=w_{21}^{KQ}=0_{d\times 1}} = \partial_{w_{21}^{KQ}}\Big[\frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2}\|w_{21}^V\|_2^2 \cdot \mathop{\mathbb{E}}_{\tau}\Big\|\begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix}^{\top}\begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^{\top} \end{pmatrix} x_{\tau,q}\Big\|_2^2\Big]\Big|_{w_{21}^V=w_{21}^{KQ}=0_{d\times 1}}$$

$$= \Big[\frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot \|w_{21}^V\|_2^2 \cdot \partial_{w_{21}^{KQ}} \mathop{\mathbb{E}}_{\tau}\Big\|\begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix}^{\top}\begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^{\top} \end{pmatrix} x_{\tau,q}\Big\|_2^2\Big]\Big|_{w_{21}^V=w_{21}^{KQ}=0_{d\times 1}}$$

$$= \frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2} \cdot \|0_{d\times 1}\|_2^2 \cdot \partial_{w_{21}^{KQ}}\Big[\mathop{\mathbb{E}}_{\tau}\Big\|\begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix}^{\top}\begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^{\top} \end{pmatrix} x_{\tau,q}\Big\|_2^2\Big]\Big|_{w_{21}^{KQ}=0_{d\times 1}} = 0_{1\times d}.$$

The above two equations justify the claim in Step 4.

Step 5: Based on results from previous Steps 1 to 4, we eventually have that

$$\partial_{w_{21}^V}\tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big|_{w_{21}^V=w_{21}^{KQ}=0_{d\times 1}} = \partial_{w_{21}^V}[\ell_1(\theta)+\ell_2(\theta)+\ell_3(\theta)+\ell_4(\theta)]\Big|_{w_{21}^V=w_{21}^{KQ}=0_{d\times 1}} = \sum_{i=1}^{4} 0_{1\times d} = 0_{1\times d},$$

$$\partial_{w_{21}^{KQ}}\tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big|_{w_{21}^V=w_{21}^{KQ}=0_{d\times 1}} = \partial_{w_{21}^{KQ}}[\ell_1(\theta)+\ell_2(\theta)+\ell_3(\theta)+\ell_4(\theta)]\Big|_{w_{21}^V=w_{21}^{KQ}=0_{d\times 1}} = \sum_{i=1}^{4} 0_{1\times d} = 0_{1\times d}.$$

The proof is completed. □

With Lemma A.6, we can then simplify the surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$, as shown in the following Lemma A.7.

Lemma A.7. *Under Assumption 1, the surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ defined in Eq. (9) can be simplified as follows,*

$$\begin{aligned}\tilde{\mathcal{L}}^{\text{adv}}(\theta) = {}& 2\text{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big] \\ & - 4\text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot\Lambda^{\frac{3}{2}}\Big] + 2\text{Tr}(\Lambda),\end{aligned}$$

where $\Gamma(M) := \frac{N+M+1}{N+M}\Lambda + \frac{\text{Tr}(\Lambda)}{N+M}I_d$ and $\psi(M) := \frac{M^2\text{Tr}(\Lambda)}{(N+M)^2}$ are same functions as that defined in Eq. (10).

Proof. When Assumption 1 holds, by applying Lemma A.6, one can substitute terms w_{21}^V and w_{21}^{KQ} in the surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ with the zero vector $0_{d\times 1}$, which thus simplifies $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ as follows,

$$\begin{aligned}\tilde{\mathcal{L}}^{\text{adv}}(\theta) = {}& 2\mathop{\mathbb{E}}_{\tau}\Big[\begin{pmatrix}0_{1\times d} & w_{22}^V\end{pmatrix}\frac{\begin{pmatrix}X_\tau & X_\tau^{\text{sfx}} & x_{\tau,q}\\ Y_\tau & Y_\tau^{\text{sfx}} & 0\end{pmatrix}\begin{pmatrix}X_\tau & X_\tau^{\text{sfx}} & x_{\tau,q}\\ Y_\tau & Y_\tau^{\text{sfx}} & 0\end{pmatrix}^\top}{N+M_{\text{train}}}\begin{pmatrix}W_{11}^{KQ}\\ 0_{1\times d}\end{pmatrix}x_{\tau,q} - y_{\tau,q}\Big]^2 \\ & + 0 + \frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2}\mathop{\mathbb{E}}_{\tau}\Big[\|W_{11}^{KQ}x_{\tau,q}\|_2^2\cdot\|\begin{pmatrix}0_{1\times d} & w_{22}^V\end{pmatrix}\begin{pmatrix}X_\tau^{\text{sfx}}\\ Y_\tau^{\text{sfx}}\end{pmatrix}\|_2^2\Big] + 0 \\ = {}& \underbrace{2\cdot\mathop{\mathbb{E}}_{\tau}\Big[w_{22}^V\cdot\frac{Y_\tau X_\tau + Y_\tau^{\text{sfx}}X_\tau^{\text{sfx}}}{N+M_{\text{train}}}\cdot W_{11}^{KQ}x_{\tau,q} - y_{\tau,q}\Big]^2}_{:=B_1(\theta)} \\ & + \underbrace{\frac{2\epsilon^2 M_{\text{train}}}{(N+M_{\text{train}})^2}\cdot\mathop{\mathbb{E}}_{\tau}\Big[\|W_{11}^{KQ}x_{\tau,q}\|_2^2\cdot\|w_{22}^V Y_\tau^{\text{sfx}}\|_2^2\Big]}_{:=B_2(\theta)}.\end{aligned} \tag{A.15}$$

For the term $B_1(\theta)$ in Eq. (A.15), we have that

$$\begin{aligned}B_1(\theta) := {}& 2\cdot\mathop{\mathbb{E}}_{\tau}\Big[w_{22}^V\cdot\frac{Y_\tau X_\tau^\top + Y_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top}{N+M_{\text{train}}}\cdot W_{11}^{KQ}x_{\tau,q} - y_{\tau,q}\Big]^2 \\ = {}& 2\cdot\mathop{\mathbb{E}}_{\tau}\Big[\frac{w_\tau^\top\cdot(X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top)}{N+M_{\text{train}}}\cdot w_{22}^V W_{11}^{KQ}\cdot x_{\tau,q} - w_\tau^\top x_{\tau,q}\Big]^2 \\ = {}& 2\cdot\mathop{\mathbb{E}}_{\tau}\Big[\Big[\frac{X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top}{N+M_{\text{train}}}\cdot w_{22}^V W_{11}^{KQ}x_{\tau,q} - x_{\tau,q}\Big]^\top\cdot w_\tau w_\tau^\top\cdot\Big[\frac{X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top}{N+M_{\text{train}}}\cdot w_{22}^V W_{11}^{KQ}x_{\tau,q} - x_{\tau,q}\Big]\Big] \\ = {}& 2\cdot\mathop{\mathbb{E}}_{\tau}\Big[\Big[\frac{X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top}{N+M_{\text{train}}}\cdot w_{22}^V W_{11}^{KQ}x_{\tau,q} - x_{\tau,q}\Big]^\top\cdot I_d\cdot\Big[\frac{X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top}{N+M_{\text{train}}}\cdot w_{22}^V W_{11}^{KQ}x_{\tau,q} - x_{\tau,q}\Big]\Big] \\ = {}& 2\cdot\mathop{\mathbb{E}}_{\tau}\Big[x_{\tau,q}^\top\cdot(w_{22}^V W_{11}^{KQ})^\top\cdot\frac{\mathbb{E}_\tau\Big[(X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top)(X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top)\Big]}{(N+M_{\text{train}})^2}\cdot w_{22}^V W_{11}^{KQ}\cdot x_{\tau,q}\Big] \\ & - 4\cdot\mathop{\mathbb{E}}_{\tau}\Big[x_{\tau,q}^\top\cdot\frac{\mathbb{E}_\tau\Big[(X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top)\Big]}{N+M_{\text{train}}}\cdot(w_{22}^V W_{11}^{KQ})\cdot x_{\tau,q}\Big] + 2\cdot\mathop{\mathbb{E}}_{\tau}\Big[x_{\tau,q}^\top\cdot x_{\tau,q}\Big].\end{aligned} \tag{A.16}$$

For $\mathbb{E}_\tau\Big[(X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top)(X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top)\Big]$ in Eq. (A.16), we have

$$
\begin{aligned}
&\mathop{\mathbb{E}}_\tau\Big[(X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top)(X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top)\Big] \\
&= \mathop{\mathbb{E}}_\tau[X_\tau X_\tau^\top X_\tau X_\tau^\top] + \mathop{\mathbb{E}}_\tau[X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top] \cdot \mathop{\mathbb{E}}_\tau[X_\tau X_\tau^\top] \\
&\quad + \mathop{\mathbb{E}}_\tau[X_\tau X_\tau^\top] \cdot \mathop{\mathbb{E}}_\tau[X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top] + \mathop{\mathbb{E}}_\tau[X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top] \\
&= \mathop{\mathbb{E}}_\tau\Big[\sum_{i,j} x_{\tau,i} x_{\tau,i}^\top x_{\tau,j} x_{\tau,j}^\top\Big] + \mathop{\mathbb{E}}_\tau\Big[\sum_i x_{\tau,i}^{\text{sfx}}(x_{\tau,i}^{\text{sfx}})^\top\Big] \cdot \mathop{\mathbb{E}}_\tau\Big[\sum_i x_{\tau,i} x_{\tau,i}^\top\Big] \\
&\quad + \mathop{\mathbb{E}}_\tau\Big[\sum_i x_{\tau,i} x_{\tau,i}^\top\Big] \cdot \mathop{\mathbb{E}}_\tau\Big[x_{\tau,i}^{\text{sfx}}(x_{\tau,i}^{\text{sfx}})^\top\Big] + \mathop{\mathbb{E}}_\tau\Big[\sum_{i,j} x_{\tau,i}^{\text{sfx}}(x_{\tau,i}^{\text{sfx}})^\top x_{\tau,j}^{\text{sfx}}(x_{\tau,j}^{\text{sfx}})^\top\Big] \\
&= \mathop{\mathbb{E}}_\tau\Big[\sum_i x_{\tau,i} x_{\tau,i}^\top x_{\tau,i} x_{\tau,i}^\top + \sum_{1\le i,j\le N, i\ne j} \Lambda^2\Big] + M_{\text{train}}\Lambda \cdot N\Lambda + N\Lambda \cdot M_{\text{train}}\Lambda \\
&\quad + \mathop{\mathbb{E}}_\tau\Big[\sum_i x_{\tau,i}^{\text{sfx}}(x_{\tau,i}^{\text{sfx}})^\top x_{\tau,i}^{\text{sfx}}(x_{\tau,i}^{\text{sfx}})^\top + \sum_{1\le i,j\le M_{\text{train}}, i\ne j} \Lambda^2\Big] \\
&= \mathop{\mathbb{E}}_\tau\Big[\sum_{i=1}^{N} \underbrace{(2\Lambda^2 + \text{Tr}(\Lambda)\Lambda)}_{\text{by Lemma A.1}}\Big] + (N^2 - N)\cdot\Lambda^2 + 2NM_{\text{train}}\cdot\Lambda^2 \\
&\quad + \mathop{\mathbb{E}}_\tau\Big[\sum_{i=1}^{M_{\text{train}}} \underbrace{(2\Lambda^2 + \text{Tr}(\Lambda)\Lambda)}_{\text{by Lemma A.1}}\Big] + (M_{\text{train}}^2 - M_{\text{train}})\cdot\Lambda^2 \\
&= (N^2 + N + M_{\text{train}}^2 + M_{\text{train}} + 2NM_{\text{train}})\cdot\Lambda^2 + (N + M_{\text{train}})\cdot\text{Tr}(\Lambda)\cdot\Lambda \\
&= (N + M_{\text{train}})\cdot((N + M_{\text{train}} + 1)\cdot\Lambda^2 + \text{Tr}(\Lambda)\cdot\Lambda) = (N + M_{\text{train}})^2\cdot\Gamma(M_{\text{train}})\Lambda. \qquad \text{(A.17)}
\end{aligned}
$$

For $\mathbb{E}_\tau\Big[X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top\Big]$ in Eq. (A.16), we have

$$
\begin{aligned}
&\mathop{\mathbb{E}}_\tau\Big[X_\tau X_\tau^\top + X_\tau^{\text{sfx}}(X_\tau^{\text{sfx}})^\top\Big] \\
&= \mathop{\mathbb{E}}_\tau\Big[\sum_i x_{\tau,i} x_{\tau,i}^\top\Big] + \mathop{\mathbb{E}}_\tau\Big[\sum_i x_{\tau,i}^{\text{sfx}}(x_{\tau,i}^{\text{sfx}})^\top\Big] \\
&= N\Lambda + M_{\text{train}}\Lambda = (N + M_{\text{train}})\cdot\Lambda. \qquad \text{(A.18)}
\end{aligned}
$$

Inserting Eqs. (A.17) and (A.18) into Eq. (A.16) leads to

$$
\begin{aligned}
B_1(\theta) &= 2\cdot\mathop{\mathbb{E}}_\tau\Big[x_{\tau,q}^\top\cdot(w_{22}^V W_{11}^{KQ})^\top\cdot\Gamma(M_{\text{train}})\Lambda\cdot w_{22}^V W_{11}^{KQ}\cdot x_{\tau,q}\Big] \\
&\quad - 4\cdot\mathop{\mathbb{E}}_\tau\Big[x_{\tau,q}^\top\cdot\Lambda\cdot w_{22}^V W_{11}^{KQ}\cdot x_{\tau,q}\Big] + 2\cdot\mathop{\mathbb{E}}_\tau\Big[x_{\tau,q}^\top\cdot x_{\tau,q}\Big]. \\
&= 2\cdot\underbrace{\text{Tr}\Big[(w_{22}^V W_{11}^{KQ})^\top\cdot\Gamma(M_{\text{train}})\Lambda\cdot w_{22}^V W_{11}^{KQ}\cdot\Lambda\Big]}_{\text{by Lemma A.2}} \\
&\quad - 4\cdot\underbrace{\text{Tr}\Big[\Lambda\cdot w_{22}^V W_{11}^{KQ}\cdot\Lambda\Big]}_{\text{by Lemma A.2}} + 2\cdot\text{Tr}(\Lambda) \\
&= 2\cdot\underbrace{\text{Tr}\Big[\Gamma(M_{\text{train}})\Lambda\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big]}_{\text{by Lemma A.3}} \\
&\quad - 4\cdot\underbrace{\text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot\Lambda^{\frac{3}{2}}\Big]}_{\text{by Lemma A.3}} + 2\cdot\text{Tr}(\Lambda). \qquad \text{(A.19)}
\end{aligned}
$$